

Shor's Algorithm, Read in the Right Basis

Phase kickback, one circuit with two descriptions, and the arithmetic that builds it

A condensed synthesis of the source notes — August 2026

The whole story in one paragraph

Multiplication by a modulo N is a permutation of residues, hence a unitary U_a , and on the orbit of 1 it is a *cyclic shift* whose length is the sought period r . A shift is diagonalised by the Fourier basis, so U_a 's eigenvalues are $e^{2\pi i s/r}$; **the period is an angle**. An eigenvalue applied to a state is a global phase — invisible — so we attach a control register: because an eigenvector is a fixed point, controlled- U_a^x leaves the work register alone and *kicks the phase back* onto the control, building the plane wave $\sum_x e^{2\pi i x s/r} |x\rangle$. We cannot prepare an eigenvector without knowing r , but $|1\rangle$ is secretly the uniform sum of all r of them, and linearity runs the circuit on every eigenbranch at once; the branches are orthogonal, so measurement conditions on one uniformly random s . The inverse QFT is a bank of matched filters that turns the ramp's frequency into a readable position $\ell \approx 2^t s/r$; continued fractions recover r ; and $\gcd(a^{r/2} \pm 1, N)$ factors N . Everything else — reversible adders, modular reduction, uncomputation — exists to build $c-U_a$ without leaving a record that would kill the interference.

Three morals organise everything below; watch them recur at every scale.

M1 — Operators are basis-free; matrices are not

An operator becomes a matrix only once you fix a basis, and every unitary owns a basis — its eigenbasis — in which its matrix is *diagonal*: nothing mixes, each axis just spins in place by a phase. A unitary is a rotation; its eigenvectors are the axes. Pick the basis that diagonalises the operator you care about and mysteries become bookkeeping.

M2 — A scalar has no address

The tensor product is bilinear: $(\lambda|a\rangle) \otimes |b\rangle = |a\rangle \otimes (\lambda|b\rangle) = \lambda(|a\rangle \otimes |b\rangle)$. A phase belongs to a whole branch of a superposition, not to either register. Phase kickback is this one line, used on branches. A phase on the *whole* state is unobservable; only phases that are *relative* between branches of some register are physical.

M3 — Interference needs indistinguishable branches; collapse is conditioning

Two computational branches add amplitudes only if they agree on *every* register; any leftover record multiplies their cross term by the overlap of the records. Measurement samples an outcome with probability $|\text{amplitude}|^2$ and then simply *conditions* the state on it. Records decide what can interfere; conditioning decides what you see.

Notation, fixed once

N : odd composite to factor, $n = \lceil \log_2 N \rceil$ bits. a : base with $\gcd(a, N) = 1$; $r = \text{ord}_N(a)$, the least $r > 0$ with $a^r \equiv 1 \pmod{N}$. t : qubits of the *counting* (control, top) register; the *work* (target, bottom) register has n qubits. Kets $|x\rangle$ for computational basis states, identified with integers; the shorthand $|a\rangle|b\rangle \equiv |a\rangle \otimes |b\rangle$ (also $|a, b\rangle$, $|ab\rangle$) denotes a state of a multi-register system. $\langle \phi | \psi \rangle$: inner product, conjugate-linear on the left. Phases are written in *cycles*, $\phi \in [0, 1)$, as $e^{2\pi i \phi}$. The Fourier transform on M points uses the positive sign, $F_M |j\rangle = M^{-1/2} \sum_k e^{2\pi i jk/M} |k\rangle$, so $F_2 = H$. $U_a |y\rangle = |ay \bmod N\rangle$; its eigenvectors are $|u_s\rangle$, $s = 0, \dots, r-1$. $c-U$ denotes controlled- U . In circuit diagrams time flows left to right, each horizontal line is *one qubit* for the whole computation — the diagram is closer to sheet music than to a plumbing schematic: nothing “flows along the wires”; a column is one instant. A slashed line labelled n is a bundle of n qubits.

1 The quantum mechanics you need (five facts)

Everything below uses exactly five facts.¹

F1 (States). The state of k qubits is a unit vector $|\psi\rangle = \sum_x \psi_x |x\rangle$ in $\mathcal{H} = (\mathbb{C}^2)^{\otimes k} \cong \mathbb{C}^{2^k}$: a table assigning a complex *amplitude* ψ_x to each of the 2^k basis states, with $\sum_x |\psi_x|^2 = 1$. States differing by a *global* phase, $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$, are the same physical state; *relative* phases between the ψ_x are physical. A qubit is not a probabilistic bit: $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$ both give 50:50 statistics in the computational basis, yet $\langle + | - \rangle = 0$ — one Hadamard ($H|\pm\rangle = |0\rangle$ resp. $|1\rangle$) turns the invisible sign into a certainty. The extra structure is the relative phase, and it is the entire resource.

F2 (Evolution: unitaries are rotations). A gate is a linear map specified by where it sends basis states, extended by linearity — a map over the (basis state, amplitude) table. It must be *unitary*, $U^\dagger U = \mathbf{1}$, equivalently $\langle U\phi | U\psi \rangle = \langle \phi | \psi \rangle$:

¹The framing of this section — a state as a table of (basis state, amplitude) pairs, gates as maps over that table “in the functional-programming sense”, and measurement as ordinary conditioning (“physicists have a flair for the dramatic”, they say the state has *collapsed*) — follows A. Albarghouthi's PLDI keynote, <https://www.youtube.com/watch?v=-zeNQ840okw>. His claim there is worth repeating: locked in a room with these facts and enough patience, you could re-derive essentially all of quantum computing. This note is an instance of that exercise.

lengths and angles are preserved, so a circuit only ever *rotates* the state vector of $\mathbb{C}^{2^k}$; it can move amplitude around but never create it. Unitarity also means invertibility — every subroutine must be reversible (§9).

F3 (Composition: tensor product). Registers combine by $\otimes$, dimensions *multiply*, and the product is *bilinear* (§3). This is where the exponential enters: $t + n$ qubits carry 2^{t+n} amplitudes. Had nature chosen the direct sum ($d_A + d_B$) instead of the product ($d_A d_B$) there would be no quantum computing — this is the observation of Manin and Feynman that started the field.

F4 (Measurement = sample, then condition). Measuring all qubits in the computational basis returns x with probability $|\psi_x|^2$ and leaves $|x\rangle$. Measuring only *register A* of a joint state is conditioning, nothing more: write

$$|\Psi\rangle = \sum_b v_b |b\rangle_A |\psi_b\rangle_B, \quad v_b \geq 0, \quad |\psi_b\rangle \text{ unit vectors};$$

outcome b occurs with probability v_b^2 (the squared norm of the whole b -component), and the post-measurement state is $|b\rangle_A |\psi_b\rangle_B$ — the surviving branch, renormalised, *with all relative amplitudes and phases inside it untouched*. “Collapse” is conditional probability with amplitudes upstream of it. Two consequences used later: measuring a subsystem destroys only which-branch information, not the coherence within a branch; and measuring in another basis $\{V|k\rangle\}$ is the same as applying $V^\dagger$ and measuring computationally — *the inverse QFT at the end of Shor is exactly this move*.

F5 (Interference — a theorem, not a postulate). If outcome x is reachable along several branches with amplitudes A_j , then

$$\Pr[x] = \left| \sum_j A_j \right|^2 = \underbrace{\sum_j |A_j|^2}_{\text{classical}} + \underbrace{\sum_{j \neq j'} \overline{A_{j'}} A_j}_{\text{cross terms}},$$

and the cross terms can be negative: paths cancel. This follows from linearity (F1–F2) plus the squaring in F4. It is the only reason quantum beats classical, and protecting these cross terms is what most of Shor’s circuit is for.

Two lemmas about unitaries carry the whole algorithm.

Lemma 1 (eigenvalues are angles). *If $U|u\rangle = \lambda|u\rangle$ with $\langle u|u\rangle = 1$, then $|\lambda| = 1$, so $\lambda = e^{2\pi i\varphi}$.*

Proof. $1 = \langle u|u\rangle = \langle u|U^\dagger U|u\rangle = |\lambda|^2$. □

An eigenvalue of a unitary carries no magnitude — only an angle. If a unitary is to tell us a number, that number *must* be encoded as a phase. This is forced, not chosen.

Lemma 2 (distinct eigenvalues, orthogonal eigenvectors). *If $U|u\rangle = \lambda|u\rangle$, $U|u'\rangle = \lambda'|u'\rangle$ with $\lambda \neq \lambda'$, then $\langle u'|u\rangle = 0$.*

Proof. $\langle u'|u\rangle = \langle u'|U^\dagger U|u\rangle = \overline{\lambda'}\lambda \langle u'|u\rangle$ and $\overline{\lambda'}\lambda \neq 1$. □

Lemma 2 is why, later, the r eigenbranches of Shor’s state can never interfere with one another, and why one measurement cleanly picks one of them.

2 Operators become matrices only in a basis

An operator is a basis-free geometric object — “rotate by π about this axis”. A *matrix* is that object written in coordinates, and it changes when the coordinates do: if V is the change of basis, the same operator has matrix A in the old basis and $V^\dagger A V$ in the new one. The smallest example already contains all of Shor:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \text{ in } \{|0\rangle, |1\rangle\}, \quad X = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \text{ in } \{|+\rangle, |-\rangle\}, \quad X = HZH.$$

In the computational basis X is a *permutation*: it swaps the two axes, and no entry of the matrix is complex — you cannot see a phase anywhere. In its own eigenbasis the same operator does not move anything: it fixes the $|+\rangle$ axis and multiplies the $|-\rangle$ axis by $-1 = e^{i\pi}$. The permutation and the phases are two descriptions of one rotation; H is the dictionary between them.

Theorem 3 (spectral theorem for unitaries). *Every unitary U on $\mathbb{C}^d$ has an orthonormal basis of eigenvectors $|u_j\rangle$ with $U|u_j\rangle = e^{2\pi i\varphi_j}|u_j\rangle$, i.e.*

$$U = \sum_j e^{2\pi i\varphi_j} |u_j\rangle\langle u_j|, \quad [U]_{\text{eig}} = \text{diag}(e^{2\pi i\varphi_1}, \dots, e^{2\pi i\varphi_d}).$$

Why diagonal? Because each eigenvector maps only to itself: $\langle u_k | U | u_j \rangle = e^{2\pi i \phi_j} \langle u_k | u_j \rangle = 0$ for $k \neq j$. **This is what “diagonal” means physically: in the eigenbasis, nothing mixes.** U decomposes into d independent one-dimensional phase rotations. Combined with F2, this is the precise sense in which *every unitary is a rotation and its eigenvectors are the axes*: the axes are the directions that do not move (they only spin in place by their eigenphase), and the whole action of U is “multiply each axis by its angle”. For one qubit this is literal geometry: up to a global phase every 2×2 unitary is a rotation $R_{\hat{n}}(\theta) = e^{-i\theta \hat{n} \cdot \vec{\sigma}/2}$ of the Bloch sphere, and its two eigenstates are the *antipodal points on the rotation axis* $\hat{n}$. (X is the π -rotation about the x -axis; its axis pierces the sphere at $|+\rangle$ and $|-\rangle$.)

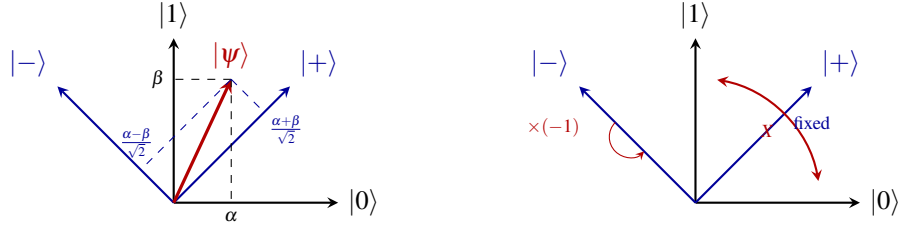


Figure 1: **One qubit, two coordinate systems** (real amplitudes drawn; the true state space is complex — for general phases the honest picture is the Bloch sphere, where $\{|\pm\rangle\}$ are the antipodes of the x -axis). *Left*: the same vector $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ has coordinates (α, β) on the computational axes and $(\frac{\alpha+\beta}{\sqrt{2}}, \frac{\alpha-\beta}{\sqrt{2}})$ on the $|\pm\rangle$ axes; H converts between them. *Right*: X described twice — it *swaps* the computational axes (a permutation, no phase in sight), while in its own eigenbasis it fixes $|+\rangle$ and multiplies $|-\rangle$ by -1 (nothing moves; one axis spins).

The same game one size up. In the computational basis CNOT is again a 0/1 permutation matrix; in the basis $\{|0\rangle, |1\rangle\} \otimes \{|+\rangle, |-\rangle\}$ — change the *target’s* coordinates only — it is diagonal:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \text{ in } \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}, \quad \text{CNOT} = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix} \text{ in } \{|0+\rangle, |0-\rangle, |1+\rangle, |1-\rangle\},$$

since $\text{CNOT}|1\rangle|-\rangle = |1\rangle X|-\rangle = -|1\rangle|-\rangle$ and the other three basis states are fixed. Read that diagonal matrix again: it is $\text{diag}(1, 1, 1, -1)$, i.e. *in the right basis, CNOT is a controlled phase* — and a diagonal gate has no preferred control wire ($\text{diag}(1, 1, 1, e^{i\phi})|x\rangle|y\rangle = e^{i\phi} |x\rangle|y\rangle$ is symmetric in $x \leftrightarrow y$, which is why such gates are drawn as two dots joined by a line). “Which register acts on which” is a property of the description, not of the operator. That is phase kickback, one section early.

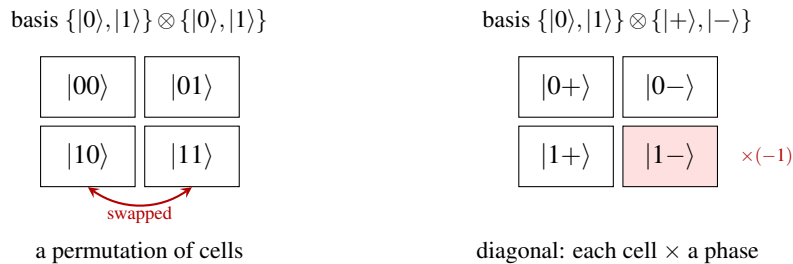


Figure 2: **Two qubits, change of basis on the target only.** The joint basis is a grid (control label $\times$ target label). CNOT permutes cells of the left grid; on the right grid it fixes every cell and multiplies $|1\rangle|-\rangle$ by -1 . Same operator, two matrices.

Superposition is relative to a basis; entanglement is relative to a cut

$|+\rangle$ is a superposition in the Z basis and a basis state in the X basis: “being in superposition” is a property of a state *together with a basis*, and can always be removed by a basis change. Entanglement (§3) is a property of a state *together with a cut* between registers, and survives every local basis change $U_A \otimes U_B$. An algorithm’s job is to arrange that the final state is *not* in superposition with respect to the measurement basis — the last gate layer (one H , or $F^\dagger$) is precisely that arranging.

3 Tensor products: where phases live

For single systems $\mathcal{H}_A = \mathbb{C}^{d_A}$, $\mathcal{H}_B = \mathbb{C}^{d_B}$, the composite space is $\mathcal{H}_A \otimes \mathcal{H}_B \cong \mathbb{C}^{d_A d_B}$, spanned by the products $|i\rangle \otimes |j\rangle$, and $\otimes$ is *bilinear*: linear in each slot separately,

$$\left(\sum_i \alpha_i |i\rangle\right) \otimes \left(\sum_j \beta_j |j\rangle\right) = \sum_{i,j} \alpha_i \beta_j |i\rangle \otimes |j\rangle, \quad \boxed{|a\rangle \otimes (\lambda |b\rangle) = (\lambda |a\rangle) \otimes |b\rangle = \lambda (|a\rangle \otimes |b\rangle)}$$

Inner products factor, $\langle a' \otimes b' | a \otimes b \rangle = \langle a' | a \rangle \langle b' | b \rangle$, and operators act factorwise, $(A \otimes B)(|a\rangle \otimes |b\rangle) = A|a\rangle \otimes B|b\rangle$; a gate on one register alone is really $A \otimes \mathbf{1}$.

Why bilinearity? It is not a new law of physics

Bilinearity is not a dynamical assumption; it is what the tensor product *is*. It is forced by consistency: superposing a subsystem must be the same as superposing the joint system ($(\alpha|\psi_1\rangle + \beta|\psi_2\rangle) \otimes |\phi\rangle$ must equal $\alpha|\psi_1\rangle \otimes |\phi\rangle + \beta|\psi_2\rangle \otimes |\phi\rangle$, or linear evolution on a part would contradict linear evolution on the whole), and it encodes that amplitudes of independent systems *multiply*, exactly as independent probabilities do. The boxed identity — **a scalar has no address** — is the single algebraic fact behind phase kickback: no phase ever “travels” between registers; the three expressions are literally the same vector, and only bookkeeping changes.

A state that can be written as one product $|\phi\rangle_A \otimes |\chi\rangle_B$ is *separable*; one that cannot is *entangled*. For two qubits, $|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$ is a product iff the coefficient matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has rank one, i.e. iff $ad - bc = 0$; in general, product $\iff$ the coefficient matrix (rows: register A’s basis, columns: register B’s) has rank one. Number of terms proves nothing — $|+\rangle|+\rangle$ has four terms and is a product; the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$ has $ad - bc = \frac{1}{2} \neq 0$: the two qubits *have no individual states*, only the pair does.

Finally, the phase dichotomy that drives the algorithm. $e^{i\gamma}|\Psi\rangle$ and $|\Psi\rangle$ give identical statistics for every measurement (F4 squares moduli), so a *global* phase is unobservable. But in $\alpha|0\rangle + \beta e^{i\theta}|1\rangle$ the phase θ is *relative* between branches and shows up in any measurement that makes the branches interfere: measuring in the $|\pm\rangle$ basis gives $p(\pm) = \frac{1}{2}(1 \pm \cos \theta)$ for the state $(|0\rangle + e^{i\theta}|1\rangle)/\sqrt{2}$. An eigenvalue relation $U|u\rangle = e^{2\pi i\phi}|u\rangle$ by itself only ever produces global phases — apply U as many times as you like to $|u\rangle$, nothing is measurable. **Apply it to one branch of a superposition, and the same number becomes relative, hence physical.** Producing that situation is exactly what a *controlled* gate does; this is why phase estimation cannot be built from uncontrolled applications of U , no matter how many.

4 Phase kickback

A controlled- U is a *joint* operator on control $\otimes$ target:

$$\text{c-}U = |0\rangle\langle 0| \otimes \mathbf{1} + |1\rangle\langle 1| \otimes U = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & U \end{pmatrix}. \quad (1)$$

Colloquially “the control acts on the target”, but (1) plays no favourites — and on the right states the information visibly flows the other way.

Theorem 4 (phase kickback). *Let $U|u\rangle = e^{2\pi i\phi}|u\rangle$. Then for any control amplitudes,*

$$\text{c-}U \left[(\alpha|0\rangle + \beta|1\rangle) \otimes |u\rangle \right] = (\alpha|0\rangle + \beta e^{2\pi i\phi}|1\rangle) \otimes |u\rangle.$$

Proof. By (1) and linearity, the state maps to $\alpha|0\rangle \otimes |u\rangle + \beta|1\rangle \otimes e^{2\pi i\phi}|u\rangle$. Move the scalar across the tensor sign (M2): $\beta|1\rangle \otimes e^{2\pi i\phi}|u\rangle = (\beta e^{2\pi i\phi}|1\rangle) \otimes |u\rangle$. Factor out $|u\rangle$. $\square$

The proof is three lines and consists entirely of moving a scalar across a tensor sign. Nothing physically travelled from target to control; what changed is which description is useful. Three things have happened at once, and they are the crux:

The three simultaneous facts

(i) **The target does not move.** An eigenvector is a fixed point of U up to a scalar; the controlled gate cannot change the work register, it can only emit a number. The target serves purely as a *phase source* — a catalyst: it enters as $|u\rangle$, leaves as $|u\rangle$, and (no-cloning notwithstanding) one copy can be reused by every gate in the circuit. (ii) **The scalar lands on the control.** Branch $|0\rangle$ got 1, branch $|1\rangle$ got $e^{2\pi i\phi}$: what would have been an unobservable global phase is now a *relative* phase between control basis states — real, interferable structure. Information flowed backwards, from target to control, opposite to the way the gate “points”. (iii) **The registers stay a product.** No entanglement was created; the work register can be ignored from now on. All the information sits in the control.

4.1 The change-of-basis view: why kickback is not a trick

Expand $U = \sum_j e^{2\pi i \varphi_j} |u_j\rangle\langle u_j|$ (Theorem 3) inside (1):

$$c-U = |0\rangle\langle 0| \otimes \mathbf{1} + \sum_j e^{2\pi i \varphi_j} |1\rangle\langle 1| \otimes |u_j\rangle\langle u_j| \quad \implies \quad [c-U]_{\{|c\rangle \otimes |u_j\rangle\}} = \text{diag}(1, \dots, 1, e^{2\pi i \varphi_1}, e^{2\pi i \varphi_2}, \dots).$$

In the basis $\{|0\rangle, |1\rangle\} \otimes \{\text{eigenvectors of } U\}$, controlled- U is diagonal — a conditional phase, nothing else. Restricted to the two-dimensional subspace $\text{span}\{|0\rangle|u\rangle, |1\rangle|u\rangle\}$ it is exactly the phase gate $\text{diag}(1, e^{2\pi i \varphi})$ *on the control*. Kickback is not a clever manoeuvre that happens to work on eigenvectors; it is the statement that a diagonal matrix multiplies each axis by its eigenvalue, read in the right coordinates. The mystery evaporates once you stop insisting on the computational basis for the target. The smallest instance is Figure 2: CNOT with target $|-\rangle$,

$$\text{CNOT: } |+\rangle|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|-\rangle \mapsto \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|-\rangle = |-\rangle|-\rangle,$$

a *bit flip on the control* produced by a gate that “acts on the target”. In the computational basis this looks like magic; in $\{|0\rangle, |1\rangle\} \otimes \{|\pm\rangle\}$ it is the diagonal matrix $\text{diag}(1, 1, 1, -1)$ doing the only thing diagonal matrices do. Shor’s $|u_s\rangle$ is the same trick with a finer protractor.

4.2 If the target is not an eigenvector: entanglement

Feed the general $|\psi\rangle = \sum_j c_j |u_j\rangle$:

$$c-U \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |\psi\rangle \right] = \frac{1}{\sqrt{2}} \sum_j c_j (|0\rangle + e^{2\pi i \varphi_j} |1\rangle) |u_j\rangle.$$

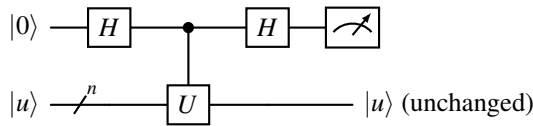
Each eigencomponent kicks back its own eigenphase. If two populated φ_j differ, the control state depends on which eigencomponent it is glued to, and the sum cannot be refactored as (control) $\otimes$ (target): trying to factor forces $e^{2\pi i \varphi_1} = e^{2\pi i \varphi_2}$ (compare coefficients, or note the coefficient matrix now has rank > 1). The registers are entangled. Quantitatively, after $c-U$ on $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |\psi\rangle$ the control alone is described by

$$\rho_{\text{ctrl}} = \frac{1}{2} \begin{pmatrix} 1 & \bar{z} \\ z & 1 \end{pmatrix}, \quad z = \langle \psi | U | \psi \rangle,$$

and an interference experiment on the control shows fringes $p(\pm) = \frac{1}{2}(1 \pm \text{Re } z)$ of visibility $|z| \leq 1$, with equality iff $|\psi\rangle$ is an eigenvector. Hence four descriptions, one fact: $|\psi\rangle$ is an eigenvector of $U = c-U$ creates no entanglement = the target records no which-path information = the fringes have full visibility. Eigenvectors are precisely the states on which a controlled gate is coherence-preserving — and when the target does record which branch the control took, the fringes wash out by exactly the recorded amount (this becomes the garbage story of §9, and, fed the state $|1\rangle = \sum_s |u_s\rangle/\sqrt{r}$, it becomes the “one random eigenphase” story of §8).

4.3 One gate yields one phase, not one bit

After kickback the control is $(|0\rangle + e^{2\pi i \varphi} |1\rangle)/\sqrt{2}$, whose computational statistics are 50:50 *for every* φ : the phase is stored where a computational measurement cannot see it. It must be followed by a basis change — one H for one qubit, $F^\dagger$ for many — and *the design of that basis change is the design of the algorithm*. The one-qubit prototype is the Mach–Zehnder interferometer in circuit form:



$p(0) = \cos^2(\pi\varphi)$, $p(1) = \sin^2(\pi\varphi)$: the first H splits the paths, $c-U$ is a phase plate in the upper arm, the second H recombines the paths and converts phase into probability.

The control cannot be dispensed with: uncontrolled U ’s on $|u\rangle$ produce only a global phase. The control is what splits the world into two branches so the phase becomes relative; building the *controlled* version of U_a is therefore a genuine engineering requirement (§10), not pedantry.

One theorem, four appearances: shifts are diagonal in the Fourier basis

Let $A_c|y\rangle = |(y+c) \bmod M\rangle$ be the shift on $\mathbb{Z}_M$. The Fourier states $|\hat{\ell}\rangle = F_M^\dagger|\ell\rangle = M^{-1/2} \sum_y e^{-2\pi i \ell y/M} |y\rangle$ are simultaneous eigenvectors of every shift:

$$A_c|\hat{\ell}\rangle = e^{2\pi i c \ell/M} |\hat{\ell}\rangle \quad \iff \quad F_M^\dagger A_c F_M \text{ is diagonal}$$

(substitute $y' = y + c$; the sum re-indexes cyclically). *Every* kickback below is this theorem at some M : $M = 2$:

$A_1 = X, F_2 = H, |\hat{1}\rangle = |-\rangle$ — the CNOT/ $|-\rangle$ example above, and the textbook oracle trick $U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$. $M = 2^n$: addition of a constant is diagonal in the Fourier basis — run in reverse, this *builds* an adder out of phase gates (Draper, §10.1). $M = r$: U_a is a cyclic shift along the orbit of 1; its eigenvectors are Fourier vectors on that cycle — Shor (§6). And the counting register at the end of Shor holds a Fourier state whose “ ℓ ” is the answer; $F^\dagger$ rotates it back to the computational basis (§5).

5 From one kickback to phase estimation

Kickback with one control qubit writes one phase; phase estimation is t kickbacks at doubling rates, then the basis change that reads them. Given $c\text{-}U^{2^j}$ for $j = 0, \dots, t-1$ and an eigenvector $U|u\rangle = e^{2\pi i\phi}|u\rangle$: put the counting register in uniform superposition, $H^{\otimes t}|0\rangle^{\otimes t} = 2^{-t/2} \sum_{x=0}^{2^t-1} |x\rangle$ (each qubit $|0\rangle \mapsto |+\rangle$; “all exponents at once”), then apply the ladder, whose joint action is the controlled power

$$\sum_x |x\rangle\langle x| \otimes U^x \quad (\text{rung } j \text{ applies } U^{2^j} \text{ conditioned on bit } x_j, \text{ and } \prod_j (U^{2^j})^{x_j} = U^{\sum_j x_j 2^j} = U^x).$$

On the eigenvector each branch $|x\rangle$ acquires $e^{2\pi i x \phi}$ (Theorem 4, t times over), and since $|u\rangle$ is common to all branches it factors out:

$$\frac{1}{2^{t/2}} \sum_x |x\rangle \otimes |u\rangle \mapsto \underbrace{\frac{1}{2^{t/2}} \bigotimes_{j=t-1}^0 (|0\rangle + e^{2\pi i 2^j \phi} |1\rangle)}_{\text{still a product state}} \otimes |u\rangle = \left[\frac{1}{2^{t/2}} \sum_{x=0}^{2^t-1} e^{2\pi i \phi x} |x\rangle \right] \otimes |u\rangle. \quad (2)$$

The counting register now carries a *phase ramp*: as x increases by 1 the phase advances by the fixed angle $2\pi\phi$ — a discrete plane wave of frequency ϕ , one bit of ϕ per control qubit. Measuring the frequency of a wave is what Fourier transforms do.

Reading the ramp. The cleanest way to see it: the ramp of frequency $\ell/2^t$ is *literally* $F_{2^t}|\ell\rangle$. So if $\phi = \ell/2^t$ exactly, $F^\dagger$ maps the counting register to $|\ell\rangle$ with certainty — no approximation, no interference argument needed. In general, by F4, “measure in the Fourier basis” = “apply $F^\dagger$, then measure computationally”: the amplitude of outcome ℓ is

$$a_\ell = \frac{1}{2^t} \sum_{x=0}^{2^t-1} e^{2\pi i x \delta}, \quad \delta = \phi - \frac{\ell}{2^t}.$$

Read this as a bank of 2^t matched filters evaluated in superposition: each ℓ is the hypothesis “the frequency is $\ell/2^t$ ”, and $F^\dagger$ ’s factor $e^{-2\pi i x \ell/2^t}$ unwinds the ramp at that trial rate. Right hypothesis: every term is 1 — 2^t unit vectors adding coherently. Wrong hypothesis: the residual phases sweep the circle and cancel. Constructive interference on the truth, destructive elsewhere.

Theorem 5 (QPE accuracy). *Let ℓ be a nearest integer to $2^t \phi$, so $|2^t \delta| \leq \frac{1}{2}$. Then $\Pr[\ell] = |a_\ell|^2 \geq 4/\pi^2 \approx 0.405$; the two nearest integers together carry $\geq 8/\pi^2 \approx 0.81$.*

Proof. For $\delta \neq 0$ sum the geometric series and use $|1 - e^{i\theta}| = 2|\sin(\theta/2)|$: $|a_\ell| = 2^{-t} |\sin(\pi 2^t \delta)| / |\sin(\pi \delta)|$. Bound the numerator below by the chord ($|\sin \pi \theta| \geq 2|\theta|$ for $|\theta| \leq \frac{1}{2}$, applied to $\theta = 2^t \delta$) and the denominator above by the arc ($|\sin u| \leq |u|$): $|a_\ell| \geq 2^{-t} \cdot \frac{2 \cdot 2^t |\delta|}{\pi |\delta|} = \frac{2}{\pi}$. $\square$

$|a_\ell|^2$ as a function of ℓ is a Dirichlet-kernel peak of width ~ 1 centred on $2^t \phi$, with $1/\ell^2$ tails; more counting qubits sharpen the estimate *additively* (each extra qubit buys one bit). One warning before using this on U_a : the rungs are $U, U^2, U^4, \dots, U^{2^{t-1}}$, and building them by repetition costs $2^t - 1$ applications — exponential. **Phase estimation is efficient only for unitaries whose powers are cheap.** For a generic black-box U there is no escape and QPE offers no speedup at all; Shor’s algorithm exists because modular arithmetic hands us cheap powers (§6) and a cheap circuit for one power (§10).

6 Shor’s operator and its spectrum

6.1 Why orders factor integers

Fix odd composite N (not a prime power — test and strip such cases classically), pick a with $\gcd(a, N) = 1$, and let $r = \text{ord}_N(a)$.

Theorem 6. *If r is even and $a^{r/2} \not\equiv -1 \pmod{N}$, then $\gcd(a^{r/2} - 1, N)$ is a nontrivial factor of N .*

Proof. Put $x = a^{r/2}$, so $N \mid x^2 - 1 = (x-1)(x+1)$. $N \nmid x-1$ (else $a^{r/2} \equiv 1$ contradicts minimality of r) and $N \nmid x+1$ (hypothesis). So the prime factors of N split between $x-1$ and $x+1$: $1 < \gcd(x-1, N) < N$. $\square$

Fact 7. For N odd with $m \geq 2$ distinct prime factors and a uniform on the units mod N , the hypotheses of Theorem 6 hold with probability $\geq 1 - 2^{-(m-1)} \geq \frac{1}{2}$ (CRT + counting 2-adic valuations of the local orders).²

Everything therefore rests on computing r — for which no efficient classical algorithm is known, and which is what the quantum circuit delivers.

6.2 From the integer a to a unitary

On n qubits define

$$U_a|y\rangle = \begin{cases} |ay \bmod N\rangle, & 0 \leq y < N, \\ |y\rangle, & N \leq y < 2^n. \end{cases}$$

Since $\gcd(a, N) = 1$, $y \mapsto ay$ is a bijection of $\{0, \dots, N-1\}$; extended by the identity it is a bijection of all 2^n labels, so U_a is a *permutation matrix* — automatically unitary. **Unitarity here is a number-theoretic condition**, and the hypothesis $\gcd(a, N) = 1$ is doing two jobs: it makes U_a a permutation at all, and it makes $\bar{a} = a^{-1} \bmod N$ exist, which is what lets the multiplier be realised *in place* (§10.4). Two consequences:

Gift 1: the powers collapse

$U_a^{2^j} = U_{a^{2^j} \bmod N}$. A high power of U_a is *not a long circuit* — it is the same circuit with a different classical constant, precomputed by repeated squaring. This turns the $2^t - 1$ applications that QPE threatens into t rungs, and it is the reason QPE applies to U_a at all. It is not routine; it is the whole trick.

Second, and this is where r enters the building: on the orbit of 1, i.e. the cycle $\{1, a, a^2, \dots, a^{r-1}\}$ of labels, U_a acts as a *cyclic shift* $a^k \mapsto a^{k+1}$, wrapping after exactly r steps. The period is present in the circuit as the length of a cycle — and by the shift theorem (end of §4), a cyclic shift is diagonalised by the Fourier basis *on that cycle*. For $s = 0, \dots, r-1$ define

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k / r} |a^k \bmod N\rangle \quad (3)$$

Lemma 8. $U_a|u_s\rangle = e^{2\pi i s / r}|u_s\rangle$, and $\langle u_s | u_{s'} \rangle = \delta_{ss'}$.

Proof. $U_a|u_s\rangle = r^{-1/2} \sum_k e^{-2\pi i s k / r} |a^{k+1}\rangle$; re-index $k' = k+1$, running cyclically since $a^r \equiv a^0$: the coefficient of $|a^{k'}\rangle$ is $e^{-2\pi i s (k'-1)/r} = e^{2\pi i s / r} e^{-2\pi i s k' / r}$. Orthonormality is the character sum $\langle u_{s'} | u_s \rangle = r^{-1} \sum_k e^{2\pi i (s'-s)k / r} = \delta_{ss'}$. $\square$

Here is M1 at Shor scale — the same operator, two matrices (shown for $r = 4$, basis $|1\rangle, |a\rangle, |a^2\rangle, |a^3\rangle$ of the orbit):

$$[U_a]_{\text{comp}} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad \text{vs.} \quad [U_a]_{\text{eig}} = \begin{pmatrix} 1 & & & \\ & i & & \\ & & -1 & \\ & & & -i \end{pmatrix}, \quad \text{entries } e^{2\pi i s / 4}.$$

Every entry on the left is 0 or 1: **the order r is invisible in the matrix entries of U_a , but it is written plainly in the eigenvalues $e^{2\pi i s / r}$** . Phase estimation is the instrument that reads eigenvalues. (Why the phases in (3) are forced: $|0\rangle$ is fixed by U_a , eigenvalue 1, useless; the *unphased* orbit sum $r^{-1/2} \sum_k |a^k\rangle$ is also eigenvalue 1, because the shift merely permutes identical terms. One must break the symmetry by attaching phases, so that the shift can no longer be absorbed — and then the eigenvalue picks up exactly one factor $e^{2\pi i s / r}$: a number containing r .)

6.3 The circular-input problem and the $|1\rangle$ trick

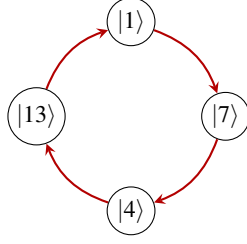
Writing down $|u_s\rangle$ requires knowing r — the very thing we seek. Phase estimation looks circular. It is not:

Lemma 9. $\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$, and more generally $|a^k \bmod N\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} e^{2\pi i s k / r} |u_s\rangle$.

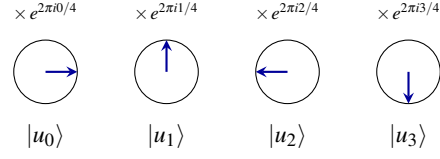
Proof. Exchange the sums: $r^{-1/2} \sum_s |u_s\rangle = r^{-1} \sum_k (\sum_s e^{-2\pi i s k / r}) |a^k\rangle$; the inner geometric series is $r \delta_{k,0}$ (ratio $z = e^{-2\pi i k / r}$: all terms 1 if $k = 0$; otherwise $z \neq 1$, $z^r = 1$, sum $(1 - z^r)/(1 - z) = 0$). The general identity is the inverse Fourier transform on the cycle. $\square$

²The bound $1 - 2^{-m}$ sometimes quoted is false; the exponent $m-1$ is correct (smallest counterexample with two distinct primes: $N = 21$, where exactly half of the good bases work).

U_7 on the orbit of 1 in $\mathbb{Z}_{15}$



computational basis: everything moves



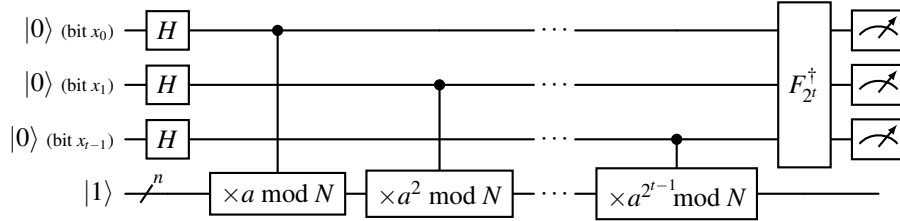
eigenbasis: nothing moves; each axis spins in place

Figure 3: U_a in two bases ($N = 15$, $a = 7$: orbit $1 \rightarrow 7 \rightarrow 4 \rightarrow 13 \rightarrow 1$, so $r = 4$). *Left*: in the computational basis U_a is a cyclic shift of the orbit — a permutation with entries 0, 1. *Right*: in the Fourier basis of the cycle, the same operator is r independent phase wheels: applying U_a multiplies the axis $|u_s\rangle$ by $e^{2\pi i s/r}$. The period is the number of notches on the wheel.

$|1\rangle = |0 \cdots 01\rangle$ costs one X gate to prepare, and it is *secretly the uniform superposition of all r eigenvectors* — the Fourier transform of a delta function is flat, transported to the orbit. The circuit is linear, so running it on $|1\rangle$ runs it on every eigenbranch simultaneously; by Lemma 2 the branches are orthogonal and cannot interfere; measurement conditions onto one of them. We will obtain an estimate of s/r for a *uniformly random* s , not chosen by us — the price of not knowing the eigenvector, and §11 pays it.

7 One circuit, two readings

Assemble the machine: Hadamards, the ladder of controlled modular multiplications, inverse QFT, measure.



Rung j multiplies the work register by the precomputed constant $a^{2^j} \bmod N$, conditioned on bit x_j (Gift 1; the rungs all commute, so their order is irrelevant). After the Hadamards the state is $2^{-t/2} \sum_x |x\rangle |1\rangle$; the ladder then acts. Here is the hinge of the whole subject — *the same middle layer, described twice*:

Reading A (computational basis — the ladder tabulates f). By Gift 1 the joint action is $|x\rangle|y\rangle \mapsto |x\rangle|a^x y \bmod N\rangle$: modular exponentiation, with *the exponent living in the top register*. On the input $|1\rangle$,

$$\frac{1}{\sqrt{2^t}} \sum_{x=0}^{2^t-1} |x\rangle |1\rangle \mapsto \frac{1}{\sqrt{2^t}} \sum_{x=0}^{2^t-1} |x\rangle |a^x \bmod N\rangle. \quad (4)$$

This state is literally a lookup table of $f(x) = a^x \bmod N$: each branch is one row, argument in the top register, value in the bottom, and crucially $|x\rangle$ is *not consumed* — the entangled state (4) is the graph of f , all 2^t rows at once. This is Shor's original 1994 description: build a periodic function in superposition, then Fourier-sample its period.

Reading B (eigenbasis — the ladder kicks phases). Substitute Lemma 9 into (4) and swap the order of summation (or: expand $|1\rangle$ first and apply kickback branchwise, $|x\rangle|u_s\rangle \mapsto e^{2\pi i x s/r} |x\rangle|u_s\rangle$):

$$\frac{1}{\sqrt{2^t}} \sum_x |x\rangle |a^x \bmod N\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \underbrace{\left[\frac{1}{\sqrt{2^t}} \sum_x e^{2\pi i x s/r} |x\rangle \right]}_{\text{phase ramp of frequency } s/r} \otimes |u_s\rangle \quad (5)$$

— one QPE phase ramp (2) per eigenbranch, tagged by the orthogonal label $|u_s\rangle$. This is Kitaev's and Cleve–Ekert–Macchiavello–Mosca's description: *phase estimation on U_a* .

Remark 10 (No gate ever writes a phase). Every gate in the ladder is a permutation matrix; below the final $F^\dagger$ there is no complex number anywhere in the circuit. The phases in (5) are not added by anything: they are what a permutation *looks*

like in its own Fourier eigenbasis, and they become physically meaningful only because the permutation was applied to a *superposition*, where a shuffle of basis labels turns into relative phases between branches. Reading A is “build a periodic table and Fourier-sample it” (Shor); Reading B is “estimate an eigenphase of U_a ” (Kitaev, CEMM). One circuit, two descriptions — not two algorithms. The phases appear on the change of basis, *and that change of basis is the algorithm*. *Remark 11* (The entanglement rank is the period). (5) is a Schmidt decomposition with r equal coefficients $1/\sqrt{r}$: the Schmidt rank of Shor’s pre-QFT state is r . The whole eigenvalue story is this decomposition read aloud.

8 Why one run yields one eigenphase

Apply $F^\dagger$ to the counting register of (5). Each ramp is (approximately) rotated onto its frequency label:

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |\widetilde{2^t s/r}\rangle \otimes |u_s\rangle,$$

where $|\widetilde{2^t s/r}\rangle$ is concentrated on the integers nearest $2^t s/r$ (exactly $|2^t s/r\rangle$ when $r \mid 2^t$). Now use the two lemmas of §1. The tags $|u_s\rangle$ are mutually orthogonal (Lemma 2), so branches with different s *never interfere* — for measurement statistics of the counting register the state behaves as a classical mixture, $\rho_{\text{count}} = \frac{1}{r} \sum_s |\widetilde{2^t s/r}\rangle \langle \widetilde{2^t s/r}|$. Measurement is conditioning (F4): the outcome is $\ell \approx 2^t s/r$ where s is uniform on $\{0, \dots, r-1\}$, and conditioned on the outcome the work register is projected into the corresponding eigenbranch. One run, one phase, one (random) eigenvector.

Contrast: a pure eigenvector in the work register. Had we fed a single $|u_s\rangle$, the joint state would remain a *product* at every step — (2) — the work register never entangles, and the readout is (near-)deterministic for that one s . With $|1\rangle$ the state (5) is entangled, and that is not a defect: the entanglement is exactly what correlates each eigenvector with its own phase estimate so that measuring the control samples the spectrum. What we lose is the choice of s ; what we keep is that *any* s with $\gcd(s, r) = 1$ determines r .

The imagined measurement that need not happen

Suppose someone measured the work register in the eigenbasis $\{|u_s\rangle\}$ *before* the circuit ran: the problem would collapse to the single-eigenvector case with s uniform. But that measurement is in an eigenbasis of U_a , so it commutes with every rung $c-U_a^{2^j}$ — it can be slid to the end of the circuit, where it manifestly cannot affect the counting-register statistics. Hence it need not be performed at all. Corollary: **the work register is never measured and never uncomputed**; it may simply be discarded. (Many expositions measure it anyway — harmless, by this argument, and never necessary.)

The comb: the same conclusion in Reading A. Condition instead on the work register in (4): outcome $|a^{x_0}\rangle$ leaves the counting register in the renormalised surviving branch

$$\frac{1}{\sqrt{K}} \sum_{j=0}^{K-1} |x_0 + jr\rangle, \quad K \approx 2^t/r$$

— a perfectly *coherent* comb of spacing r and unknown offset x_0 (F4: conditioning destroys only which-branch information, not coherence within the branch). Under $F^\dagger$ the offset contributes only the phase $e^{-2\pi i \ell x_0 / 2^t}$ per outcome — modulus one, gone at the squaring — so the distribution peaks at multiples of $2^t/r$ regardless of x_0 : the unknown offset is harmless, and the output distribution is *identical* to Reading B’s (r Dirichlet peaks of weight $1/r$ at $\ell \approx s 2^t/r$). Which register carries “the answer” is again a matter of description.

A common misreading

The pre-measurement state contains all r eigenphases, but this is not “computing all eigenvalues at once”: one measurement returns *one* of them, sampled with probability $1/r$ (in general $|c_s|^2$). What makes Shor work is not massive parallelism but that any single good sample suffices — and that interference concentrated the probability on the good outcomes before the sample was taken.

9 Garbage, which-path records, and uncomputation

The work register’s record $|a^x \bmod N\rangle$ was benign — its orthogonality structure (equal iff $x \equiv x' \bmod r$) is *the answer*. Real arithmetic circuits, however, threaten to keep other records, and those are fatal. Here is the whole story in one formula. Let the state before the final basis change be $|\Psi\rangle = \sum_x \alpha_x |x\rangle_A |g_x\rangle_B$ (A : the register we will transform and measure; B : everything else — scratch, work, environment), and apply $V (= F^\dagger)$ to A alone. The probability of outcome y is the squared *norm* of the whole y -component (F4):

$$P(y) = \left\| \sum_x V_{yx} \alpha_x |g_x\rangle \right\|^2 = \sum_{x, x'} V_{yx} \overline{V_{yx'}} \alpha_x \overline{\alpha_{x'}} \langle g_{x'} | g_x \rangle. \quad (\star)$$

The register you are not measuring casts its vote: every cross term is multiplied by the overlap of the records.

Case 1: no branch-dependent record ($\langle g_x \rangle = |g\rangle$ for all x ; it factors out): $P(y) = |\sum_x V_{yx} \alpha_x|^2$ — the sum sits *inside* the modulus, cross terms alive, full interference. **Case 2: a perfect record** ($\langle g_x | g_x \rangle = \delta_{xx}$): $P(y) = \sum_x |V_{yx} \alpha_x|^2$ — the sum has moved *outside* the modulus; probabilities add where amplitudes should have. In between, the fringe visibility is exactly $|\langle g_{x'} | g_x \rangle|$. The underlying criterion: amplitudes belong to complete configurations of the whole machine, so two branches share an amplitude — and can add or cancel — *exactly when they agree on every register, measured or not*.

Smallest instance (interference destroyed by one CNOT). Take $|0\rangle \xrightarrow{H} \xrightarrow{\text{CNOT to a helper}} \xrightarrow{H}$, i.e. $\alpha_0 = \alpha_1 = \frac{1}{\sqrt{2}}$, $V = H$:

$$\text{clean } (g_0 = g_1): \quad P(0) = \left| \frac{1}{2} + \frac{1}{2} \right|^2 = 1, \quad P(1) = 0; \quad \text{copied } (\langle g_1 | g_0 \rangle = 0): \quad P(0) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} = P(1).$$

Certainty becomes a coin flip. Nothing was measured and nothing was lost: one CNOT onto a helper made the branches disagree somewhere, and that was enough. This is the double slit with a which-path detector: the fringes vanish even if nobody ever reads the detector, because interference requires the two paths to end in the same total state of the world.

Shor's version. Reversible arithmetic built naively computes $|x\rangle|1\rangle|0\cdots 0\rangle \mapsto |x\rangle|a^x\rangle|g_x\rangle$, where g_x collects carry chains, comparison bits, partial products — and depends on x , because the carries depend on what was added. Then

$$P(\ell) \propto \sum_{x,x'} e^{2\pi i(x-x')\ell/2^t} \langle a^{x'} | a^x \rangle \langle g_{x'} | g_x \rangle.$$

The middle factor is the one the algorithm wants: it confines interference to $x \equiv x' \pmod{r}$, and those surviving cross terms are how the period imprints itself on the statistics. The last factor is the tax: if the scratch distinguishes x from x' *within* a residue class — and intermediate carries essentially always do — then $\langle g_{x'} | g_x \rangle = 0$, every cross term dies, and the diagonal alone gives $P(\ell) = 2^{-t}$ for every ℓ : **exactly uniform noise**. The circuit is still perfectly unitary; it simply computes nothing. (Debugging corollary: the failure mode of a dirty ancilla is not a wrong answer but a flat histogram — and a classical post-processor that verifies candidates can occasionally “succeed” on white noise, masking the bug. Assert that ancillas end in $|0\rangle$; test the distribution, not the factors.) So the question is never “is there a record?” but “*does the record separate branches I still need to add?*” — the work register separates only different residues (good: that is period finding); garbage separates everything (fatal).

Why the only exit is uncomputation. Erasing scratch ($|g_x\rangle \mapsto |0\rangle$ for all x) is many-to-one — not unitary. Discarding it changes nothing — the record still exists in the world. The remaining move is the one reversibility itself provides: every step that *wrote* the garbage is invertible, so run those steps backwards and *unwrite* it.

Theorem 12 (Bennett; compute–copy–uncompute). *If f has a reversible circuit $C : |x\rangle|0\rangle_{\text{scr}} \mapsto |x\rangle|f(x), g_x\rangle_{\text{scr}}$ of size T , then there is a circuit of size $2T + O(|f(x)|)$ realising the clean map*

$$|x\rangle|y\rangle|0\rangle_{\text{scr}} \mapsto |x\rangle|y \oplus f(x)\rangle|0\rangle_{\text{scr}},$$

with every ancilla returned to $|0\rangle$, hence unentangled from everything, in every branch.

Proof. Run C ; copy the answer with CNOTs, $|y\rangle \mapsto |y \oplus f(x)\rangle$ (fan-out of a basis-state value is not cloning); run $C^\dagger$. The ancilla state is now $|0\rangle$ independently of x , so it factors out of any superposition and every overlap $\langle g_{x'} | g_x \rangle$ in $(\star)$ returns to 1: Case 1 restored. $\square$

Two refinements matter for Shor. *In-place variant:* the map $|x\rangle \mapsto |f(x)\rangle$ (input gone) is impossible in general — the copy of x stays entangled and cannot be erased — *unless f is efficiently invertible:* compute $f(x)$ into a fresh register, then uncompute x by running the circuit for f^{-1} backwards. Shor's multiplier does exactly this with f^{-1} = multiplication by $\bar{a} = a^{-1} \pmod{N}$ (extended Euclid — the second job of $\text{gcd}(a, N) = 1$). *The price is structural:* roughly half of every arithmetic circuit below exists to run the other half backwards. That is not waste; it is the price of the interference — when $F^\dagger$ finally asks the branches to recombine, nothing anywhere in the machine may remember which branch was which.

10 Building controlled- U_a : the arithmetic tower

Phase estimation does not accept just any circuit that computes $ay \pmod{N}$. Four properties are load-bearing:

Demand	Why	Discharged by
Unitary	QPE's whole framework is spectral; no unitarity, no eigenphases	$\text{gcd}(a, N)=1 \Rightarrow$ permutation
Cheap powers	else the ladder costs $2^t - 1$ applications (§5)	Gift 1: $U_a^{2^j} = U_{a^{2^j} \pmod{N}}$
Controllable	uncontrolled U_a emits only a global phase (§3)	control threaded through every level
Garbage-free	scratch entangles and kills the interference (§9)	uncomputation at every level

The construction is a tower: the ladder is made of controlled multipliers, multipliers of modular adders, modular adders of plain adders, adders of Toffolis. Every gate below the final $F^\dagger$ is a permutation matrix — classical reversible arithmetic that happens to act linearly on superpositions.

About the code

Per request, the Python/Qiskit listings below are *illustrative and deliberately unchecked*: they document the structure (block $\leftrightarrow$ theorem correspondence; little-endian convention, $q[0]$ = least significant bit), not a tested implementation.

10.1 Addition the Fourier way (Draper)

The shift theorem (§4) at $M = 2^n$, read as a design principle: *adding a constant is a shift, and shifts are diagonal in the Fourier basis* — so transform, apply phases, transform back. Concretely, the QFT factors (y's bits little-endian),

$$F_{2^n}|y\rangle = \bigotimes_{i=0}^{n-1} \frac{|0\rangle + e^{2\pi i y 2^{i-n}}|1\rangle}{\sqrt{2}},$$

so the classically known constant X is added by n *uncontrolled single-qubit phase gates*: qubit i gets $P(2\pi X 2^{i-n})$, turning $e^{2\pi i y 2^{i-n}}$ into $e^{2\pi i (y+X) 2^{i-n}}$, i.e. $F|y\rangle \mapsto F|y+X \bmod 2^n\rangle$. Subtraction is the same gates with $-X$ (phases are periodic, so $-X \equiv 2^n - X$); the composite $F^\dagger(\text{phases})F$ is exactly a 0/1 permutation again — the complex numbers live only between the transforms.

$$|y\rangle \xrightarrow{F} \boxed{P(2\pi X 2^{i-n}) \text{ on each qubit } i} \xrightarrow{F^\dagger} |y+X \bmod 2^n\rangle$$

```
from math import pi, gcd
from fractions import Fraction
from random import randrange
from qiskit import QuantumCircuit, QuantumRegister, ClassicalRegister

def qft(qc, q):
    # q[0] = least significant bit
    n = len(q)
    for i in reversed(range(n)):
        qc.h(q[i])
        for j in range(i):
            qc.cp(2*pi / 2.0**(i - j + 1), q[j], q[i])
    for i in range(n // 2):
        qc.swap(q[i], q[n - 1 - i])

def phi_add(qc, q, X):
    # inside Fourier space: F|y> -> F|y+X>
    for i in range(len(q)):
        qc.p(2*pi * X * 2.0**(i - len(q)), q[i]) # 2.0**: keep floats

def add_const(qc, q, X):
    # |y> -> |y + X mod 2^n>
    qft(qc, q); phi_add(qc, q, X); inverse_qft(qc, q) # qft run backwards
```

Why not stop here? Two reasons, both about scale. The smallest rotation used is $2\pi/2^n$ — at $n = 2048$ about 10^{-616} radians, far below any physical or synthesis precision (in practice one truncates to an *approximate* QFT). Worse, fault-tolerantly the rotations $R_k = \text{diag}(1, e^{2\pi i/2^k})$ for $k \geq 4$ cannot be synthesised *exactly* from Clifford+ T gates, only approximated at a cost per rotation — whereas Toffoli and CNOT are exact, and the T -count is what magic-state distillation actually bills you for. Hence the workhorse is classical-style ripple-carry arithmetic:

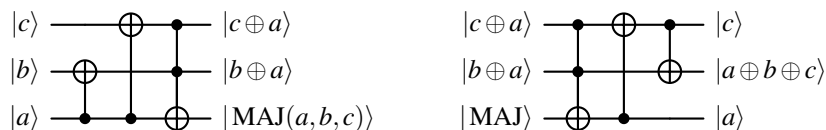
10.2 Addition the ripple way (CDKM)

Adding two n -bit registers columnwise: with carry-in $c_0 = 0$,

$$s_i = a_i \oplus b_i \oplus c_i, \quad c_{i+1} = \text{MAJ}(a_i, b_i, c_i) = a_i b_i \oplus a_i c_i \oplus b_i c_i.$$

Read this as a cost statement: the sum bits are $\mathbb{F}_2$ -linear given the carries, so they are CNOTs and essentially free; the carries are *quadratic*, so each costs a Toffoli; and they form a chain, so the depth is $\Theta(n)$. Linear parts free, quadratic parts expensive, chain gives depth — that one line prices Shor's algorithm. The Cuccaro–Draper–Kutin–Moulton adder does it in place with *one* ancilla, out of two three-gate blocks on a wire triple (c, b, a) :

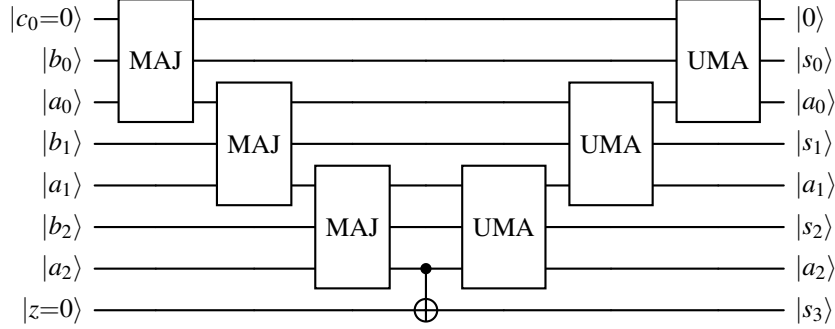
MAJ : CNOT($a \rightarrow b$), CNOT($a \rightarrow c$), TOF($c, b \rightarrow a$); UMA : TOF($c, b \rightarrow a$), CNOT($a \rightarrow c$), CNOT($c \rightarrow b$).



Lemma 13 (MAJ/UMA are partial inverses). $\text{MAJ} : (c, b, a) \mapsto (c \oplus a, b \oplus a, \text{MAJ}(a, b, c))$ and $\text{UMA} \circ \text{MAJ} : (c, b, a) \mapsto (c, a \oplus b \oplus c, a)$.

Proof. Only the Toffoli is nontrivial: it sends $a \mapsto a \oplus (c \oplus a)(b \oplus a) = ab \oplus ac \oplus bc$ over $\mathbb{F}_2$ (using $a^2 = a, a \oplus a = 0$). UMA’s Toffoli, applied to $\text{MAJ}(a, b, c)$ with the same controls, restores a ; the two CNOTs then restore c and deposit $b \oplus a \mapsto (b \oplus a) \oplus c = s$. $\square$

Chain the MAJ blocks so the carry rides up the a -wires, copy out the one clean carry, then run the UMA chain back down:



Theorem 14 (CDKM adder). *The circuit above implements $|c_0=0\rangle|b\rangle|a\rangle|z\rangle \mapsto |0\rangle|(a+b) \bmod 2^n\rangle|a\rangle|z \oplus c_n\rangle$ using $2n$ Toffolis, $4n+1$ CNOTs and $2n+2$ qubits, with the operand a preserved, addition in place on b , and zero garbage. (Fault-tolerantly $\approx 4n$ T gates: each carry is a temporary AND, uncomputed for free.)*

The design realises §9 exactly: the carries *must* be erased, so a reverse pass is compulsory — and **UMA erases the carry and writes the sum bit in the same three gates**. The paper analogy is precise: adding by hand you scribble little carry digits above the columns; here you go back over the columns rubbing out each scribble, and the act of rubbing out deposits the answer. You are not paying twice; you are paying once and collecting on the way out. Two free corollaries: running the circuit backwards subtracts, and running the MAJ chain on $a, \bar{b}$ (bitwise complement) with carry-in 1 computes $a + \bar{b} + 1 = a - b + 2^n$, whose carry-out is the predicate $c_n = [a \geq b]$ — *comparison costs nothing beyond the subtraction you were doing anyway*. That single sign bit is the workhorse of the next level.

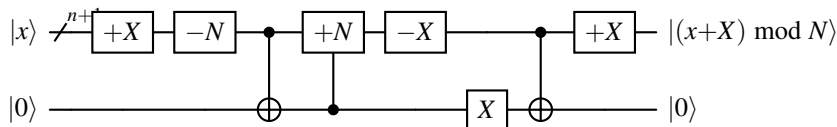
```
def maj(qc, c, b, a):
    qc.cx(a, b); qc.cx(a, c); qc.ccx(c, b, a)    # (c,b,a) -> (c^a, b^a, MAJ)

def uma(qc, c, b, a):
    qc.ccx(c, b, a); qc.cx(a, c); qc.cx(c, b)    # undo carry, deposit sum

def cdkm_add(qc, a, b, c0, z):
    # |a>|b> -> |a>|(a+b) mod 2^n>, carry -> z
    n = len(a)
    triples = [(c0, b[0], a[0])] + [(a[i-1], b[i], a[i]) for i in range(1, n)]
    for t in triples: maj(qc, *t)
    qc.cx(a[n-1], z)    # the one clean, unmodified carry
    for t in reversed(triples): uma(qc, *t)
```

10.3 Modular addition of a constant (Beauregard)

We need $x \mapsto (x + X) \bmod N$, which classically is “add; if the result is $\geq N$, subtract N ” — and a quantum computer cannot do *if*: it holds a superposition in which some branches overshot and some did not. The conditional must become arithmetic: compute the condition *into a qubit*, use that qubit as a control, and then — because the flag qubit is garbage in the sense of §9 — destroy the evidence by *recomputing the flag from the answer*. Seven blocks, on an $(n+1)$ -qubit value register (one extra top bit so intermediate values up to $2N$ fit; its top bit is the “sign” after a subtraction), plus one flag ancilla:



block	operation	value register	flag
(1)	ADD(X)	$x + X$ ($< 2N$, fits)	0
(2)	SUB(N)	$x + X - N$; sign bit = $[x + X < N]$	0
(3)	CNOT sign $\rightarrow$ flag	unchanged	$[x + X < N]$
(4)	ADD(N) if flag	$R := (x + X) \bmod N$	unchanged
(5)	SUB(X)	$R - X$; sign bit = $[R < X]$	unchanged
(6)	X on flag; CNOT sign $\rightarrow$ flag	unchanged	0
(7)	ADD(X)	R	0

Blocks (1)–(4) compute the answer; **blocks (5)–(7) exist entirely to destroy the evidence.** The flag can be cleared because it is *recomputable from the output*: since $x < N$,

$$\text{flag} = 1 \iff x + X < N \iff R = x + X \geq X, \quad \text{flag} = 0 \iff R = x + X - N < X,$$

so $[R < X] = \neg \text{flag}$; block (5) recomputes exactly that predicate into the sign bit, and (6) XORs the flag against its own recomputation. You cannot remember whether you borrowed during a subtraction — but you can look at your answer, deduce that you must have, and then honestly forget. *Every block in this tower ends by recomputing its own flag from its own output; that single discipline is what keeps the construction garbage-free.* (The precondition $x < N$ is load-bearing: fed $x \geq N$ the circuit leaves the flag dirty — and by §9 a dirty flag produces not a wrong answer but *no* answer.) One optimisation worth naming: when the whole block is itself controlled (next level), only blocks (1), (5), (7) need the extra control — a gate needs a control only if leaving it uncontrolled would fail to cancel out, and with the control off, blocks (2)(3)(4)(6) conspire to do nothing. Measured effect: an order of magnitude, from one lemma. The adders inside may be Draper’s (then the whole block lives in Fourier space between two transforms) or CDKM’s — two factorisations of one permutation.

```
def c_mod_add(qc, ctls, q, flag, X, N):
    """|x> -> |x + X mod N> on the (n+1)-qubit register q, controlled on ctls.
    add/sub = (controlled) constant adders of 10.1 or 10.2; q[-1] = sign."""
    add(qc, q, X, ctls); sub(qc, q, N)          # (1)(2) sign = [x+X < N]
    qc.cx(q[-1], flag)                         # (3)
    add(qc, q, N, [flag])                      # (4) R = (x+X) mod N
    sub(qc, q, X, ctls)                        # (5) sign = [R < X]
    qc.x(flag); qc.cx(q[-1], flag)            # (6) flag -> 0
    add(qc, q, X, ctls)                        # (7) restore R
```

10.4 Multiplication, and the in-place trick

Schoolbook multiplication is repeated shifted addition, and each shifted multiple is a *classically known* constant:

$$ax \bmod N = \left(\sum_{i=0}^{n-1} x_i (2^i a \bmod N) \right) \bmod N,$$

so the data bit x_i becomes a *control* — the data has turned into conditioning:

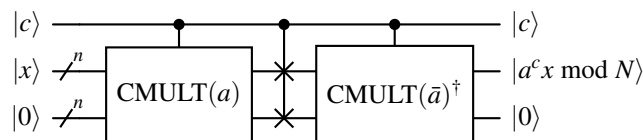
$$\text{CMULT}(a) : |c\rangle|x\rangle|b\rangle \mapsto |c\rangle|x\rangle|b + c \cdot ax \bmod N\rangle = \prod_{i=0}^{n-1} \text{ADD}_{c, x_i}^{\bmod N}(2^i a \bmod N).$$

(Precomputation gives you *one* operand of the multiplication; the other is a register in superposition over all 2^t exponents — which is exactly why the machine has to exist at all.) But CMULT writes into a *fresh* register, and the ladder must multiply the *same* register repeatedly. The out-of-place map must become $|x\rangle \mapsto |ax \bmod N\rangle$, and the leftover x must disappear — which, by §9, means *computed away*, not discarded. The escape is that a is invertible:

Theorem 15 (in-place controlled multiplier). *Let $\bar{a} = a^{-1} \bmod N$ (classical, extended Euclid). Then*

$$c\text{-}U_a = \text{CMULT}(\bar{a})^\dagger \cdot c\text{-}\text{SWAP} \cdot \text{CMULT}(a) \quad \text{implements} \quad |c\rangle|x\rangle|0\rangle \mapsto |c\rangle|a^c x \bmod N\rangle|0\rangle.$$

Proof. For $c = 1$: $|x\rangle|0\rangle \xrightarrow{\text{CMULT}(a)} |x\rangle|ax\rangle \xrightarrow{\text{SWAP}} |ax\rangle|x\rangle \xrightarrow{\text{CMULT}(\bar{a})^\dagger} |ax\rangle|x - \bar{a}(ax)\rangle = |ax\rangle|0\rangle$, since $\bar{a}a \equiv 1$. For $c = 0$ every factor is the identity. Both branches leave the scratch in $|0\rangle$, unentangled, so $c\text{-}U_a$ is realised *exactly*. $\square$



Note the dagger is cheap: the adjoint of “add $\bar{a}x$ ” is “subtract $\bar{a}x$ ”, i.e. the same adder with negated constants — adjoints of permutations are permutations, and for adders they are again adders. *The dagger becomes a minus sign on a classical number.*

```
def c_mult_acc(qc, c, x, b, flag, A, N):      # |b> -> |b + c*A*x mod N>
    for i in range(len(x)):
        c_mod_add(qc, [c, x[i]], b, flag, (A << i) % N, N)

def c_ua(qc, c, x, b, flag, A, N):          # |x>|0> -> |A^c x mod N>|0>
    c_mult_acc(qc, c, x, b, flag, A, N)
    for i in range(len(x)):
        qc.cswap(c, x[i], b[i])
    c_mult_acc(qc, c, x, b, flag, (-pow(A, -1, N)) % N, N)
    # CMULT(a^-1)^dagger: dagger = minus sign on the classical constant
```

10.5 The ladder, and the bill

Rung j of the QPE ladder is $c-U_{a^{2^j} \bmod N}$ with constants $c_0 = a$, $c_{j+1} = c_j^2 \bmod N$ precomputed classically (Gift 1) — so the compiled circuit is *specific to a* , and a failed run with a bad base means recompiling: $O(n^3)$ classical bit operations (milliseconds) against days of quantum runtime, expected about two distinct circuits in total. The zoom, and the cost it implies:

level	object	made of	multiplier	
5	the ladder	$t \approx 2n$ rungs, one per counting qubit	$\times 2n$	
4	$c-U_{a^{2^j}}$	$2 \times \text{CMULT} + n$ controlled swaps		
3	$\text{CMULT}(k)$	n doubly-controlled modular adders	$\times n$	$\underbrace{2n}_{\text{rungs}} \cdot \underbrace{2n}_{\text{adds}} \cdot \underbrace{5}_{\text{Toffoli}} \cdot \underbrace{2n}_{\text{Toffoli}} \approx 40n^3$
2	$\text{ADD}^{\bmod N}(k)$	5 constant additions + flag logic	$\times 5$	
1	$\text{ADD}(k)$	MAJ chain, CNOT, UMA chain	$\times 2n$ Toffoli	
0	Toffoli	Clifford + T	$7T; 4T$ if uncomputed	

Three of those steps each multiply by roughly n — rungs, bits of the multiplicand, bits per addition — and that, nothing subtler, is why the algorithm is cubic. At $n = 2048$ the modular exponentiation is $\sim 10^9$ – 10^{10} Toffolis against $\sim 10^5$ gates for the (approximate) Fourier layer: **the layer that carries all the conceptual content is about 0.003% of the circuit.** Everything that determines whether RSA falls is the cost of multiplying two 2048-bit numbers, reversibly, in superposition.

```
def order_circuit(A, N, t):
    n = N.bit_length()
    ctr, tgt = QuantumRegister(t, "ctr"), QuantumRegister(n, "tgt")
    anc, out = QuantumRegister(n + 2, "anc"), ClassicalRegister(t, "out")
    qc = QuantumCircuit(ctr, tgt, anc, out)
    qc.x(tgt[0])                # work register := |1>
    qc.h(ctr)                   # all exponents at once
    for j in range(t):
        c_ua(qc, ctr[j], tgt, anc[:n], anc[n], pow(A, 2**j, N), N)
    inverse_qft(qc, ctr)
    qc.measure(ctr, out)
    return qc
```

11 Classical readout: continued fractions

The measurement returns ℓ with $\left| \frac{\ell}{2^t} - \frac{s}{r} \right| \leq \frac{1}{2^{t+1}}$ with probability $\geq 4/\pi^2$ (Theorem 5). To identify the unknown fraction s/r from its approximation we need quadratic precision:

Theorem 16 (Legendre). *If $\left| \xi - \frac{p}{q} \right| < \frac{1}{2q^2}$ with $\gcd(p, q) = 1$, then p/q is a convergent of the continued-fraction expansion of ξ .*

Since $r < N$, it suffices that $2^{-(t+1)} < \frac{1}{2N^2}$, i.e. $2^t \geq 2N^2$: **take $t = 2n + 1$** . (The requirement is imposed by the classical post-processing — the quantum circuit does not care how large t is; each extra qubit is one more rung.) The recovery is then mechanical: expand $\ell/2^t$ by the Euclidean algorithm, list the $O(n)$ convergents, take the one with the largest denominator $q < N$ within $1/2q^2$, and *verify* the candidate by checking $a^q \equiv 1 \pmod{N}$ by repeated squaring. Failure modes, all cheap: ℓ falls off the peak (prob. $\leq 1 - 4/\pi^2$ per run); $s = 0$ tells nothing (prob. $1/r$); $\gcd(s, r) = d > 1$ yields the proper divisor r/d — a good s (coprime to r) occurs with probability $\phi(r)/r = \Omega(1/\log \log r)$, and two runs already recover $r = \text{lcm}(q_1, q_2)$ with constant probability. Expected $O(\log \log N)$ repetitions overall; a bad base a (Fact 7, prob. $\leq \frac{1}{2}$) restarts the outer loop.

The complete algorithm

Input odd composite N , not a prime power (test classically; strip factors of small primes).

1. Pick $a \in \{2, \dots, N-1\}$ at random; if $\gcd(a, N) > 1$, return that factor (lucky).
2. *Quantum*: run the order-finding circuit ($t = 2n+1$ counting qubits, work register $|1\rangle$, ladder of $c-U_{a^{2^j}}$, $F^\dagger$, measure) to obtain ℓ .
3. Continued fractions on $\ell/2^t \rightarrow$ candidate r ; verify $a^r \equiv 1 \pmod{N}$, else re-run / combine two runs by lcm.
4. If r odd or $a^{r/2} \equiv -1$: back to step 1. Else return $\gcd(a^{r/2} - 1, N)$ (Theorem 6).

Total cost: $O(n^3)$ gates (the arithmetic), $O(n)$ qubits, constant expected repetitions — against $\exp(O(n^{1/3} \log^{2/3} n))$ for the best known classical algorithm.

Worked micro-example: $N = 15, a = 7, t = 11$

Orbit of 1: $1 \rightarrow 7 \rightarrow 4 \rightarrow 13 \rightarrow 1$, so $r = 4$ (Fig. 3); eigenphases $s/4$. Since $r \mid 2^t$, the ramps are exact Fourier states and the output is *exactly* $\ell \in \{0, 512, 1024, 1536\}$, each with probability $\frac{1}{4}$. Say $\ell = 1536$: $1536/2048 = 3/4 = [0; 1, 3]$, convergents $0, 1, \frac{3}{4} \Rightarrow r = 4$; verify $7^4 = 2401 \equiv 1$. Then $7^2 = 49 \equiv 4 \not\equiv -1$, so $\gcd(4-1, 15) = 3$ and $\gcd(4+1, 15) = 5$: $15 = 3 \times 5$. The outcome $\ell = 1024 \rightarrow \frac{1}{2} \Rightarrow$ candidate 2, rejected by the verifier ($7^2 \equiv 4$) — the coprimality rate $\phi(4)/4 = \frac{1}{2}$, live. ($N = 15$ is atypically kind: r divides 2^t , so there are no Dirichlet tails; the honest tests are $N = 21, 33$, where the peaks smear onto neighbours and continued fractions earn their keep.)

```
def order_from_counts(counts, t, A, N):
    for y in sorted(counts, key=counts.get, reverse=True):
        y = int(y, 2)
        if y == 0: continue
        r = Fraction(y, 2**t).limit_denominator(N - 1).denominator
        if pow(A, r, N) == 1:
            # verify; limit_denominator IS the
            # continued-fraction expansion
            return r

def find_factor(N):
    n = N.bit_length()
    while True:
        A = randrange(2, N)
        if gcd(A, N) > 1: return gcd(A, N)
        r = order_from_counts(run(order_circuit(A, N, 2*n + 1)), 2*n + 1, A, N)
        if r and r % 2 == 0 and pow(A, r//2, N) != N - 1:
            d = gcd(pow(A, r//2, N) - 1, N)
            if 1 < d < N: return d
```

12 Epilogue: one theorem four times, and why it is hard to build

The document used one linear-algebra fact at four scales — *shifts are diagonal in the Fourier basis*: at $M = 2$ it was $X = HZH$ and the CNOT/ $|-\rangle$ kickback; at $M = 2^n$, run in reverse, it built the Draper adder out of phase gates; at $M = r$ it handed us U_a 's spectrum $e^{2\pi i s/r}$ and the eigenvectors $|u_s\rangle$; and at $M = 2^t$ the final $F^\dagger$ rotated the phase ramp back onto the computational basis. And it used the three morals at every joint: M1 — the ladder is a permutation in one basis and a phase generator in another, and the change of basis *is* the algorithm; M2 — every phase was moved by sliding a scalar across a tensor sign, and only phases relative between branches were ever measurable; M3 — the algorithm is one long fight to keep branches indistinguishable (uncomputation, flag re-derivation, in-place multipliers) until the one moment interference is spent, after which measurement is mere conditioning. In the closing image of the source notes: unitaries are rotations; eigenvectors are the axes; U_a 's axes write the answer as an *angle*, and $F^\dagger$ is the final rotation that converts that angle into a *position* you can read.

Why it is hard to build is the same statement read in reverse. The output depends on relative phases across $2^t \geq N^2$ branches staying coherent through $\sim 40n^3$ Toffolis. Any stray coupling to an environment, $\sum_x \psi_x |x\rangle |E_0\rangle \rightarrow \sum_x \psi_x |x\rangle |E_x\rangle$, is $(\star)$ with the environment as the register you failed to uncompute: every cross term is taxed by $\langle E_{x'} | E_x \rangle$, and complete decoherence flattens the output to exactly 2^{-t} — uniform noise. Decoherence is not noise added on top of a good answer; it is the continuous erasure of the only feature the algorithm produces. The feature that makes Shor work and the feature that makes it hard to build are the same one, which is why the machine must be fault-tolerant (error rates $\sim 10^{-11}$ per gate needed vs. $\sim 10^{-3}$ physical; error-corrected Toffolis paid for in magic states) before it can be useful. Fault tolerance is the price of using a phase as a data register.

tensor products $\rightarrow$ controlled operations $\rightarrow$ eigenvectors and kickback $\rightarrow$ superpositions of eigenvectors $\rightarrow$ entanglement $\rightarrow$
which-path records $\rightarrow$ interference and uncomputation $\rightarrow$ phase estimation $\rightarrow$ Shor.

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